Faculty of Computer Applications

INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

Introduction To Data Analytics

Healthcare Analytics And Patient Intelligence



Master of Computer Applications

Submitted To:

Mr. Divyank Chauhan

Assistant Professor

Faculty of CSED

Submitted By:

Team Name: 03 **MCA AI ML**

Group Leader: Devang Bhardwaj

Total Member: 10

Invertis University, Bareilly [UP]

(2026-27)

DECLARATION

I, Devang Bhardwaj, hereby declare that the project entitled "Healthcare Analytics Dashboard" submitted for the Master of Computer Applications (MCA) degree is my original work and has been carried out under the guidance of Mr. Divyank Chauhan, Assistant Professor, Department of Computer Applications.

The work embodied in this project has not been submitted elsewhere for the award of any other degree or diploma.

All sources of information have been duly acknowledged. The project work was completed as a collaborative effort with my team members, and each member contributed equally to its successful completion.

Team Members' Contribution:

S. No.	Student Name	Contribution
1	Devang Bhardwaj (Leader)	100%
2	Divyansh Pandey	100%
3	Ayush Saxena	100%
4	Nuris Hussain	100%
5	Mayank Pandey	100%
6	Shivam Gangwar	100%
7	Kamalpreet Kaur	100%
8	Semon Aftab	100%
9	Gajendra Kumar	100%
10	Aman Yadav	100%

Place:

Date:

Signature of Team Leader
MCA AI ML

.....

CERTIFICATE

This is to certify that the project entitled "Healthcare Analytics Dashboard" is a bonafide work carried out by Devang Bhardwaj(Team Leader) along with the team members of MCA AI ML, during the academic year 2026-2027, in partial fulfillment of the requirements for the award of Master of Computer Applications (MCA) degree.

The project work has been carried out under my guidance and supervision. The team demonstrated excellent teamwork, technical skills, and dedication throughout the project duration.

I am satisfied with the quality of work and recommend this project for evaluation.

_____

Mr. Divyank Chauhan

Assistant Professor

Project Guide

External Examiner: _____

Date: _____

Signature: _____

ACKNOWLEDGMENT

We would like to express our sincere gratitude to all those who have contributed to the successful completion of this project on "Healthcare Analytics Dashboard."

First and foremost, we extend our heartfelt thanks to our esteemed guide, Mr. Divyank Chauhan, Assistant Professor, Department of Computer Applications, for his invaluable guidance, constant encouragement, and insightful suggestions throughout the project. His expertise and mentorship were instrumental in shaping this project.

We are deeply grateful to Dr. Akash Sanghi , Head of the Department of Computer Applications, for providing us with the necessary facilities and resources to undertake this project.

We would also like to thank our college management for creating an environment conducive to learning and research, which enabled us to explore innovative ideas and implement them effectively.

Our sincere appreciation goes to all faculty members of the Department of Computer Applications who directly or indirectly helped us during the course of this project with their valuable inputs and suggestions.

We are thankful to our family members and friends for their unwavering support and encouragement, which motivated us to put in our best efforts.

Last but not least, we acknowledge the collaborative spirit of our team members. Each team member contributed wholeheartedly, bringing diverse skills and perspectives that enriched the project. The synergy and coordination among the team made this project a rewarding learning experience.

We hope this project contributes to the field of healthcare data analytics and serves as a useful reference for future research.

Team Members:
Devang , Divyansh , Ayush , Nuris , Shivam , Mayank , Kamalpreet , Semon , Gajendra , Aman

MCA AI ML

ABSTRACT

Project Title: Healthcare Analytics Dashboard
Domain: Healthcare Data Analytics, Business Intelligence
Technology Stack: Microsoft Excel, Data Visualization, Statistical Analysis

This project presents the development of a Healthcare Analytics Dashboard using Microsoft Excel. The main objective of this project is to convert raw patient data into meaningful visual insights that support better decision-making in hospitals and healthcare facilities. In the healthcare industry, data is often stored manually or in basic systems that focus only on record keeping. These traditional methods lack analytical capabilities and make it difficult to identify trends related to patients, diseases, treatment costs, and hospital performance.

To address this gap, an interactive dashboard was designed using Excel tools such as Power Query, Pivot Tables, Charts, Slicers, and KPI visualizations. The data used for the dashboard was synthetically generated using Excel formulas and Power Query randomization. The dashboard provides insights such as patient count, gender distribution, age group analysis, disease distribution, hospital-wise performance, total treatment cost, and identification of critical cases. The dashboard automatically updates based on filter selections, enabling real-time interaction and exploration of data.

The developed solution demonstrates how Excel can be used not only for data storage but also as a powerful analytical tool. The outcome of this project highlights that interactive dashboards can significantly reduce manual effort, improve reporting efficiency, and support data-driven decisions in healthcare management.

Keywords: Healthcare Analytics, Dashboard, Data Visualization, Power Query, Pivot Table, Excel, KPI.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION **7**
1.1 Background 7
1.2 Motivation 7
1.3 Problem Statement 7
1.4 Objectives 7
1.5 Scope of the Project 8
1.6 Project Overview 8

CHAPTER 2: LITERATURE REVIEW **9**
2.1 Existing System Analysis 9
2.2 Research Gap 9

CHAPTER 3: METHODOLOGY **10**
3.1 Data Collection 10
3.2 Data Cleaning & Preprocessing 10
3.3 Data Modeling 10
3.4 Dashboard Designing Process 10

CHAPTER 4: SYSTEM DESIGN **11**
4.1 Tools & Technologies Used 11
4.2 Data Flow Diagram
4.3 Architecture of Dashboard 11

CHAPTER 5: IMPLEMENTATION **12**
5.1 KPI Setup 12
5.2 Hospital Operations Dashboard 12
5.3 Patient Analytics Dashboard 12
5.4 Disease Analytics Dashboard 13
5.5 Doctor Performance Dashboard 14
5.6 Cost & Revenue Dashboard 15

CHAPTER 6: RESULTS & ANALYSIS **17**
6.1 Dashboard Output Screenshots 17
6.2 Insights / Observations 18

CHAPTER 7: CONCLUSION & FUTURE SCOPE **19**
7.1 Conclusion 19
7.2 Limitations 19
7.3 Future Scope 19

REFERENCES **20**

ANNEXURES / APPENDIX **21**
Appendix A: Data Dictionary 21
Appendix B: Python Notebook Structure 21

CHAPTER 1: INTRODUCTION

1.1 Background

In today's digital era, the healthcare sector generates a massive volume of data every day. Hospitals maintain records of patients including demographics, gender, disease type, ward, admission status, billing cost and treatment outcomes. Although the data is available, most hospitals struggle to analyze it effectively due to the lack of a proper visualization system.

Data Analytics and Business Intelligence tools transform raw data into meaningful insights. A visual dashboard helps healthcare administrators monitor activity, identify disease trends, understand patient categories and evaluate resource utilization.

This project builds **Healthcare Analytics Dashboards** using Microsoft Excel, Power BI and Python (Jupyter Notebook) on a hospital dataset of **1,000 patients**, spread over five Excel files that are linked through a common Patient ID.

1.2 Motivation

Hospital data is often stored in spreadsheets, but without analytics, valuable information remains unused. This motivated the development of dashboards that:

- Simplify data interpretation
- Reduce manual reporting time
- Provide meaningful visual insights for better decisions

Working on this project also helped enhance skills in Excel, Power BI, Python and data visualization, which are in high demand in the industry.

1.3 Problem Statement

Hospitals collect patient data, but lack an analytical system to convert that data into useful information. Major problems identified are:

- Data is scattered across several Excel sheets and records
- No automated way to track disease patterns and revenue
- Lack of comparison between wards, doctors, age groups and cities
- Manual reporting consumes time and may lead to errors

Thus, there is a need for a system that visualizes patient data effectively and presents insights to hospital management.

1.4 Objectives

The main objectives of this project are:

1. To convert raw healthcare data into visual and meaningful insights.
2. To build interactive dashboards in Excel, Power BI and Jupyter Notebook.
3. To analyze data by ward, admission status, gender, age group, city, disease and doctor.
4. To display KPIs such as total patients, total revenue, average treatment cost, average length of stay and critical cases.
5. To support healthcare decision-making through graphical representation.

1.5 Scope of the Project

The scope of the project includes:

- Importing patient data from five Excel files
- Cleaning, checking and combining the data
- Building pivot tables and summary sheets for analysis
- Designing five topic-wise dashboards and one Major Dashboard

The dashboards can be used by hospitals, healthcare management teams and research and analytics departments to monitor hospital performance and patient statistics efficiently.

1.6 Project Overview

Raw patient data was collected and processed using Excel. Five analysis areas were created, each with its own Excel file and dashboard:

- **Hospital Operation:** ward utilization, admission status, length of stay
- **Patient Analytics:** gender, age group, city
- **Disease Analytics:** disease-wise patients and severity
- **Doctor Performance:** patients treated and revenue per doctor
- **Cost and Revenue:** revenue by payment mode, disease and ward

The same data was also analysed in a Python Jupyter Notebook and in a Power BI Major Dashboard. Charts such as Pie, Donut, Column and Bar charts were used, and slicers were added in Excel for interactive filtering.

CHAPTER 2: LITERATURE REVIEW

2.1 Existing System Analysis

In most healthcare organizations, patient information is recorded and stored using paper-based registers, manual files or basic Excel sheets. These systems focus on data entry and storage rather than extracting insights. Due to the absence of analytics features, administrators struggle to understand patient trends, identify disease patterns or evaluate hospital performance.

Parameter	Existing System Description
Data Handling	Manual record-keeping or Excel sheets without automation
Reporting	Manual report preparation, time-consuming and prone to errors
Data Analysis	Very limited analytics; only basic filtering or sorting
Decision Making	Based on assumptions rather than data insights
Accessibility	Data stored in isolated files, difficult to track centrally

Problems identified:

1. **No real-time insights** – data is stored but not analyzed automatically.
2. **Slow decision-making** – management struggles to find important information quickly.
3. **Inability to track trends** – ward-wise, age-wise and disease-wise analysis is not possible.
4. **High dependency on humans** – reports need manual compilation from multiple files.
5. **Risk of errors** – manual entry increases the chance of data inaccuracy.

2.2 Research Gap

Data storage systems exist, but most do not focus on analytics and visualization. Excel and hospital management systems are widely used but are limited to record-keeping.

Existing Tool	Observed Gap
Excel	Stores data and basic operations; no automated dashboards
Hospital Management Systems	Focus on registration only, not analytics
ERP / Management software	Operational use only, no disease trend analysis
Manual reporting	Time-consuming and error-prone
Lack of monitoring tools	No insight into critical conditions and cost

A modern dashboard-based solution must convert raw data into visual charts, reduce human effort, provide interactive filters and KPIs, and improve data-driven decisions. Therefore, this project bridges the gap by providing interactive Excel, Power BI and Python dashboards that transform static patient data into dynamic visual insights.

CHAPTER 3: METHODOLOGY

3.1 Data Collection

The dataset consists of **1,000 patient records** (Patient ID P0001 to P1000) stored in five Microsoft Excel files. Admission dates range from 28 Oct 2025 to 28 Oct 2026. Each file has a *Raw Data* sheet and summary sheets.

File	Columns
1_Hospital_Operation	Patient ID, Admission Date, Discharge Date, Length of Stay, Ward Type, Admission Status
2_Patient_Analytics	Patient ID, Age, Age Group, Gender, City
3_Disease_Analytics	Patient ID, Disease, Admission Status, Age Group, Gender
4_Doctor_Performance	Patient ID, Doctor Name, Disease, Ward Type, Treatment Cost (INR)
5_Cost_and_Revenue	Patient ID, Disease, Ward Type, Treatment Cost (INR), Payment Mode

The dataset was synthetically generated (not real patient records), so no personal or medical data of any real person is used.

3.2 Data Cleaning & Preprocessing

The data was checked for accuracy and consistency before building the dashboards.

Step	Action Performed
Missing values	Checked all five files: 0 missing values
Duplicates	Patient ID is unique (1,000 rows in every file)
Data types	Dates converted to date format; age and cost kept numeric
Standardization	Gender, ward, status and payment mode values are consistent
Cross-file checks	Disease, ward, cost, gender and admission status match across files for every Patient ID

After these checks, the dataset was ready for analysis and dashboard creation.

3.3 Data Modeling

Pivot Tables were used in Excel to summarize each dataset (for example patients by ward, gender, disease and revenue by payment mode). Slicers were connected to the pivot tables for dynamic filtering. In Python, all five files were merged on **Patient ID** into one master dataset with 1,000 rows, which feeds the Major Dashboard.

The pivot tables and summary tables act as the analytical engine of every dashboard.

3.4 Dashboard Designing Process

Once the summary tables were ready, charts were added to build each dashboard.

Step	Description
Chart Selection	Pie, Donut, Bar and Column charts chosen to match the data
KPI Display	Total patients, revenue, average stay, critical cases etc.
Slicer Integration	Slicers added in Excel to filter the charts
Layout Design	Charts positioned in a clean, structured grid

CHAPTER 4: SYSTEM DESIGN

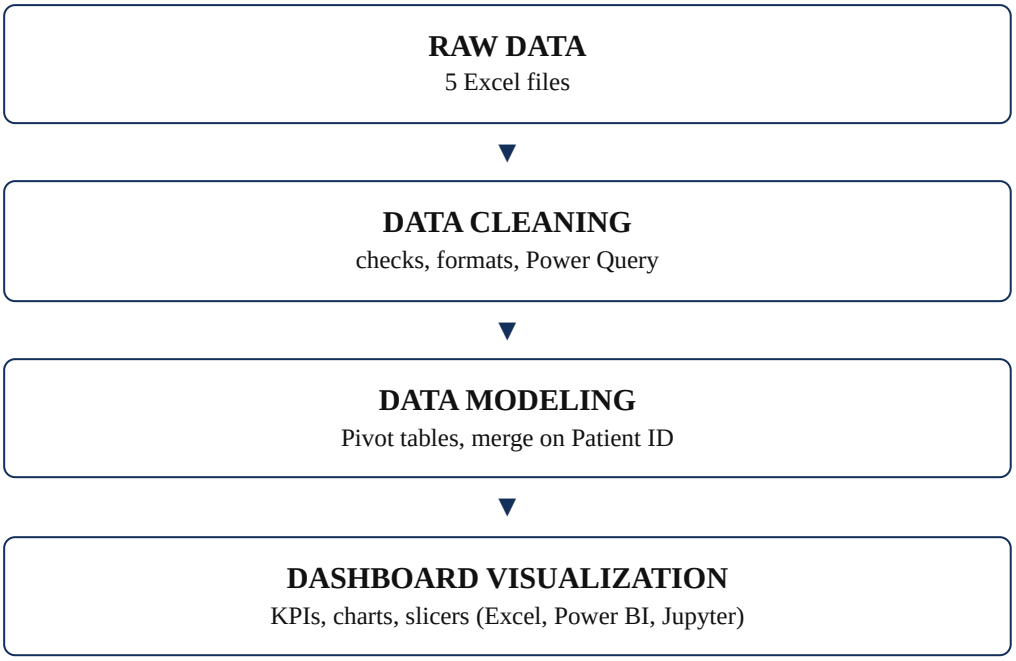
4.1 Tools & Technologies Used

The dashboards were developed using the following tools.

Tool / Technology	Purpose / Function
Microsoft Excel	Main platform: data storage, pivot tables, charts and slicers
Power Query / Pivot Tables	Clean, group and summarize the patient data
Power BI	Major Dashboard with slicers and interactive visuals
Python (pandas, matplotlib)	Data merging, analysis and chart dashboards
Jupyter Notebook	Step-by-step code and dashboard output

4.2 Data Flow Diagram (DFD)

The workflow converts raw data into visual analytics:



4.3 Architecture of Dashboard

The architecture defines how components interact to convert raw data into a visual dashboard.

Layer	Component	Function
Data Layer	Excel files (Raw Data sheets)	Stores raw patient data
Processing Layer	Power Query, Pivot Tables, pandas	Cleans data and creates summaries
Visualization Layer	Charts, KPI cards, slicers	Turns analysis into interactive insights

CHAPTER 5: IMPLEMENTATION

5.1 KPI Setup

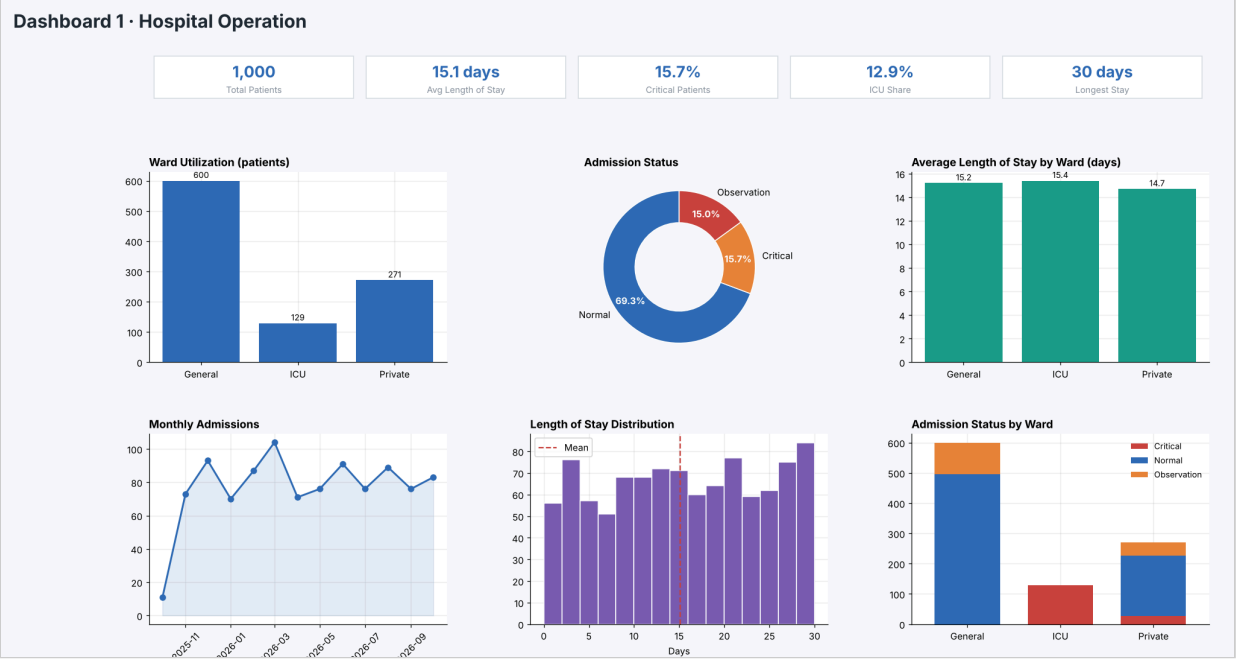
The first implementation step was to identify the most important Key Performance Indicators.

KPI	Value
Total Patients	1,000
Total Revenue	₹39.42 Cr
Average Cost per Patient	₹3.94 Lakh
Average Length of Stay	15.1 days
Critical Patients	157 (15.7%)
Average Age	43.3 years

KPIs were created using pivot tables and Excel formulas (COUNT, SUM, AVERAGE, COUNTIF).

5.2 Hospital Operations Dashboard

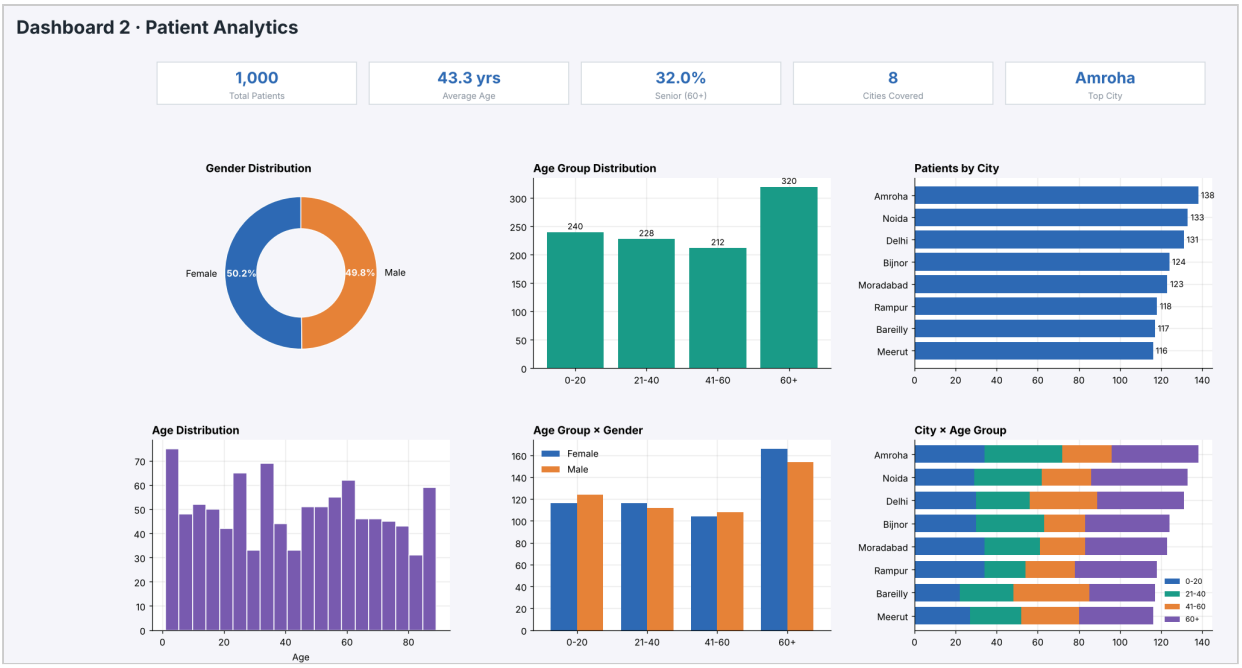
- Ward utilization, admission status and average stay were analysed.
- **General ward** has the most patients (600, 60%), followed by Private (271) and ICU (129).
- **Normal 693**, Critical 157 and Observation 150 patients.
- Average stay is about 15 days and almost the same in all wards (General 15.2, ICU 15.4, Private 14.7).



Dashboard 1: Hospital Operation (Python / Jupyter Notebook)

5.3 Patient Analytics Dashboard

- Gender is balanced: Female 502, Male 498.
- **60+** is the largest age group with 320 patients (32%); average age is 43.3 years.
- Patients come from 8 cities; Amroha (138), Noida (133) and Delhi (131) lead.

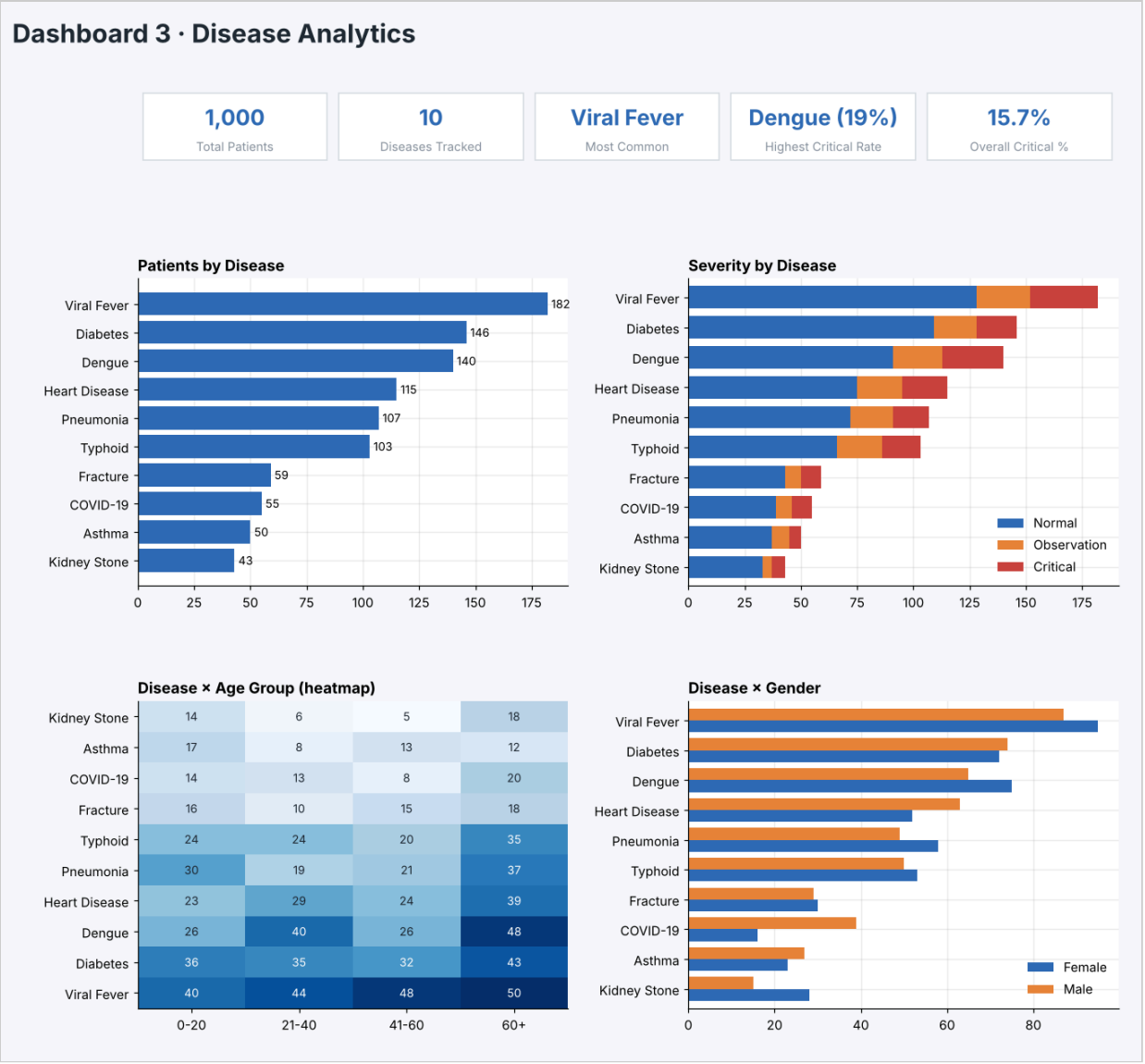


Dashboard 2: Patient Analytics (Python / Jupyter Notebook)

5.4 Disease Analytics Dashboard

Ten diseases are tracked. Viral Fever is the most common (182 patients), followed by Diabetes (146) and Dengue (140). Dengue has the highest share of critical cases (19.3%).

Disease	Patients	Critical %	Revenue
Viral Fever	182	16.5%	₹7.93 Cr
Diabetes	146	12.3%	₹5.33 Cr
Dengue	140	19.3%	₹5.98 Cr
Heart Disease	115	17.4%	₹4.61 Cr
Pneumonia	107	15.0%	₹3.90 Cr
Typhoid	103	16.5%	₹3.91 Cr
Fracture	59	15.3%	₹2.29 Cr
COVID-19	55	16.4%	₹2.49 Cr
Asthma	50	10.0%	₹1.52 Cr
Kidney Stone	43	14.0%	₹1.46 Cr



Dashboard 3: Disease Analytics (Python / Jupyter Notebook)

5.5 Doctor Performance Dashboard

Six doctors treated all 1,000 patients. Dr. Singh (Pulmo) generated the highest revenue, while Dr. Verma (Physician) treated the most patients but has the lowest average revenue per patient.

Doctor	Patients	Revenue	Avg per patient
Dr. Singh (Pulmo)	166	₹8.08 Cr	₹4.87 Lakh
Dr. Sharma (Cardio)	161	₹6.99 Cr	₹4.34 Lakh
Dr. Gupta (Ortho)	156	₹6.51 Cr	₹4.17 Lakh
Dr. Ali (Pediatrics)	154	₹6.16 Cr	₹4.00 Lakh
Dr. Verma (Physician)	189	₹6.00 Cr	₹3.18 Lakh
Dr. Khan (General)	174	₹5.68 Cr	₹3.26 Lakh



Dashboard 4: Doctor Performance (Python / Jupyter Notebook)

5.6 Cost & Revenue Dashboard

Total revenue is ₹39.42 Cr with an average bill of ₹3.94 Lakh (highest single bill ₹54.06 Lakh). Insurance is the largest payment mode.

Payment mode	Patients	Revenue	Share
Insurance	391	₹15.20 Cr	38.6%
Cash	306	₹11.19 Cr	28.4%
UPI	207	₹8.51 Cr	21.6%
Credit Card	96	₹4.52 Cr	11.5%

Ward	Patients	% patients	Revenue	% revenue	Avg bill
General	600	60.0%	₹6.85 Cr	17.4%	₹1.14 Lakh
Private	271	27.1%	₹7.38 Cr	18.7%	₹2.72 Lakh
ICU	129	12.9%	₹25.20 Cr	63.9%	₹19.53 Lakh

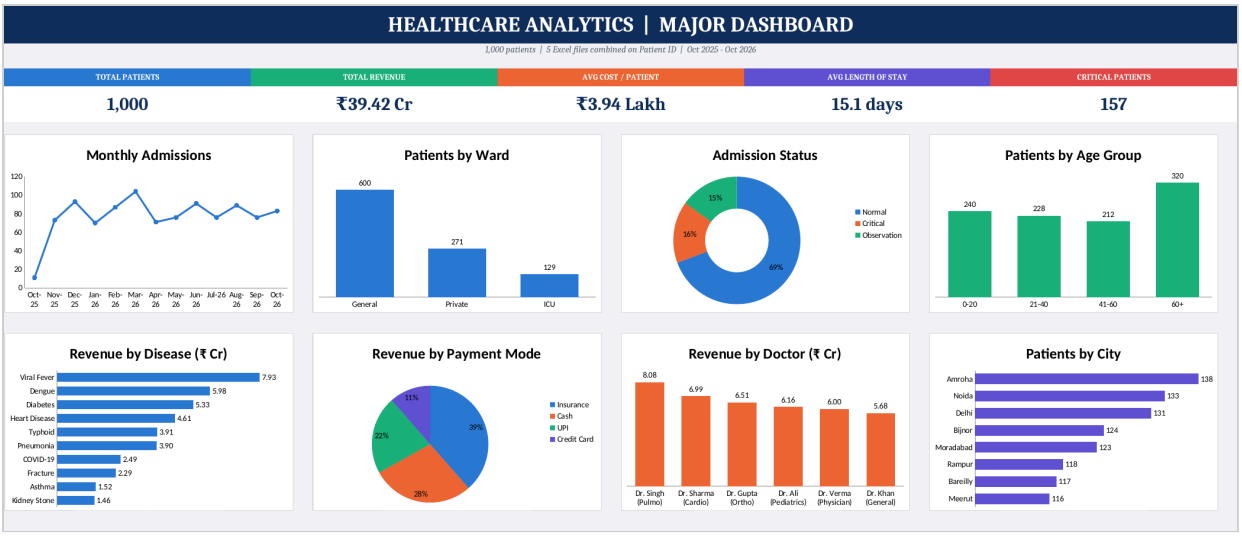


Dashboard 5: Cost & Revenue (Python / Jupyter Notebook)

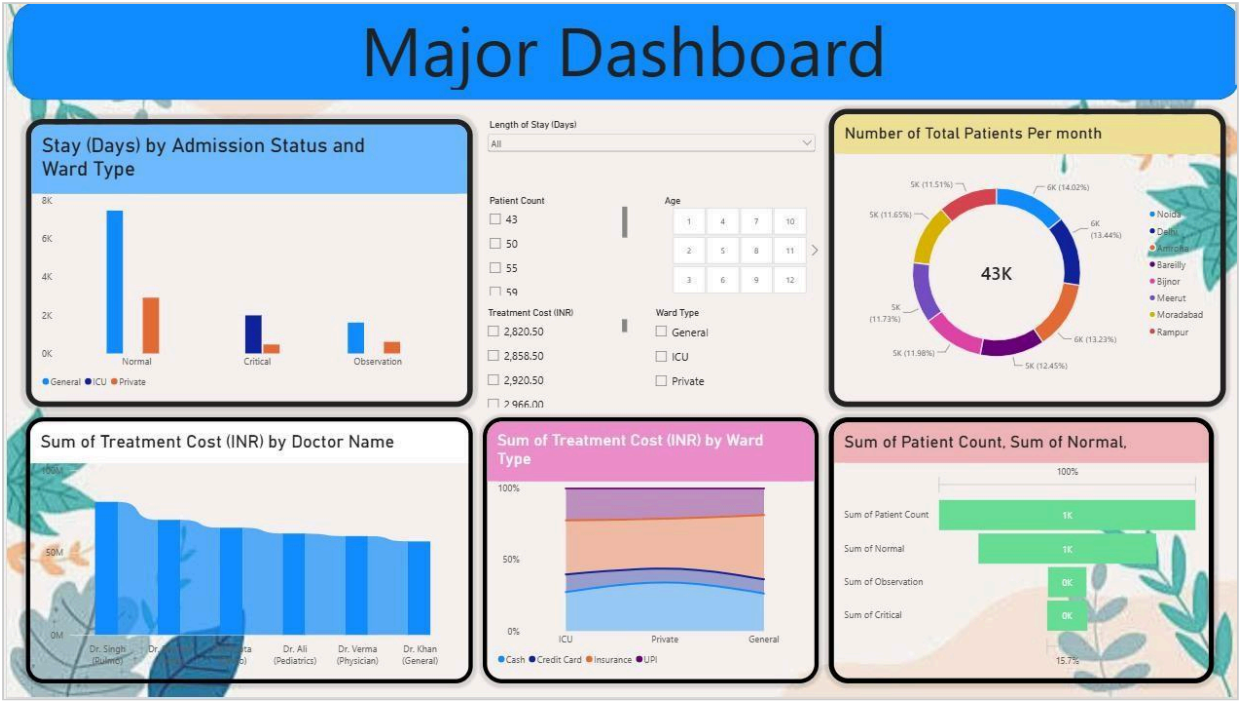
CHAPTER 6: RESULTS & ANALYSIS

6.1 Dashboard Output Screenshots

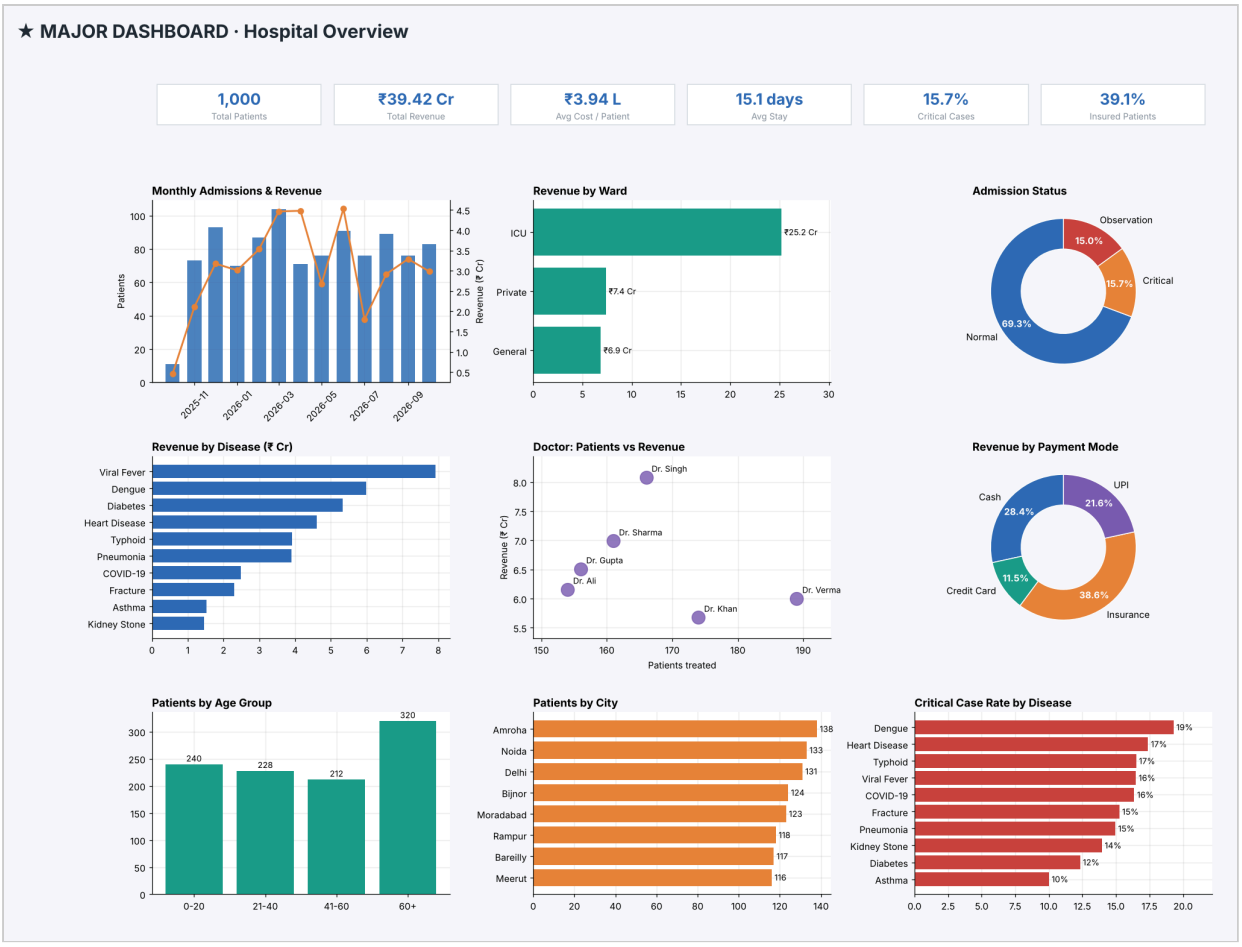
After successful implementation, the dashboards provide a clean and interactive view of patient data. The five Excel dashboards use KPI cards, pivot charts and slicers. The Major Dashboards combine all five datasets into one page.



(A) Excel Major Dashboard: KPIs and 8 charts built from the combined data



(B) Power BI Major Dashboard with slicers



(C) Python / Jupyter Notebook Major Dashboard

6.2 Insights / Observations

Observation	Description
1. Gender Distribution	Almost equal: Female 502 and Male 498.
2. Disease Trend	Viral Fever (182), Diabetes (146) and Dengue (140) are the most common.
3. Age Group Analysis	The 60+ group is the largest (32%).
4. Ward Utilization	General ward holds 60% of patients; ICU only 12.9%.
5. Cost Analysis	ICU gives 63.9% of revenue (avg bill ₹19.53 Lakh vs ₹1.14 Lakh in General).
6. Payment Mode	Insurance contributes the most revenue (38.6%).
7. Doctors	Dr. Singh earns the most revenue (₹8.08 Cr); Dr. Verma treats the most patients (189).
8. Critical Cases	157 patients (15.7%) are critical; all ICU patients are critical.

Key Takeaways:

- The dashboards help visualize overall hospital and patient performance.
- Insights can be used by administrators for better planning and resource allocation.
- The same model can be extended with real data for advanced analytics in the future.

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The Healthcare Analytics project shows how raw patient data can be converted into meaningful insights using Excel, Power BI and Python. Five topic-wise dashboards and a Major Dashboard make it easy to analyze patient count, disease distribution, age and gender, ward utilization, doctor performance, and treatment cost.

The dataset used is synthetically generated, but the results show how such dashboards can work with real hospital datasets. This project proves that Excel and Python can be used not only for data entry but also for powerful visual analytics.

7.2 Limitations

Limitation Area	Description
Synthetic Dataset	The data is generated and does not represent real medical records.
Limited Automation	Data refresh is manual in Excel.
Scalability	Excel may slow down with very large datasets.
No Predictive Analytics	Only descriptive analysis; no forecasting or machine learning.
User Access Control	Excel files cannot be shared and edited by many users at the same time.

7.3 Future Scope

Enhancement	Description
Real Patient Records	Connect to a real hospital database (SQL / cloud).
Power BI Service / Tableau	Real-time dashboards with automatic refresh.
Predictive Analytics	Predict disease trends, cost and bed utilization using machine learning.
Web Access	Host the dashboard for hospital administration teams.

REFERENCES

1. Microsoft Documentation. *Get Started with Power Query in Excel*. https://support.microsoft.com/excel
2. Microsoft Documentation. *Using Pivot Tables for Data Analysis*. https://support.microsoft.com/pivottable
3. Microsoft Learn. *Excel Charts and Visualization Tools*. https://learn.microsoft.com/excel
4. Microsoft Learn. *Power BI documentation*. https://learn.microsoft.com/power-bi
5. pandas documentation. https://pandas.pydata.org/docs
6. Matplotlib documentation. https://matplotlib.org/stable
7. IBM. *Data Analytics in Healthcare: Benefits and Use Cases* (2023). https://www.ibm.com/topics/healthcare-analytics
8. Textbook: *Data Analytics for Beginners*, Pearson Education, 2022.

ANNEXURES / APPENDIX

Appendix A: Data Dictionary

Column	Meaning
Patient ID	Unique ID (P0001 to P1000)
Admission Date / Discharge Date	Dates of admission and discharge
Length of Stay (Days)	Days between admission and discharge (0 to 30)
Ward Type	General, Private or ICU
Admission Status	Normal, Critical or Observation
Age / Age Group	Age in years; groups 0-20, 21-40, 41-60, 60+
Gender / City	Female or Male; 8 cities
Disease	10 diseases
Doctor Name	6 doctors with specialty
Treatment Cost (INR)	Cost of treatment, treated as revenue
Payment Mode	Cash, Credit Card, Insurance or UPI

Appendix B: Python Notebook Structure

- Setup: load all five Excel files with pandas and define chart helper functions.
- Dashboards 1 to 5: one dashboard per Excel file (KPI strip + six charts each).
- Major Dashboard: merge all files on Patient ID, run six consistency checks, then build 6 KPIs and 9 charts.
- Run order: place the five Excel files next to the notebook and use Kernel → Restart & Run All.